Evidence Based Resources

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Introduction:

➢ Before making an important decision, an evidence-based practitioner starts by asking, What is the available evidence Instead of basing a decision on personal judgment alone, an evidence based practitioner finds out what is known by looking for evidence from multiple sources.

➢ According to the principles of evidence base practice, There are literally millions of published reports, journal articles, correspondence and studies available to clinicians.

➢ Choosing the best resource to search is an important decision. A number of skills and actions are necessary for achieving this step, including acquiring a basic set of skills and obtaining access to a number of resources.

❖ **Definition of Evidence-based resources (EBRs):**

- Are reviews of published intervention evaluations or studies that have Evidence of effectiveness, feasibility, reach, sustainability, and transferability.

☒ **Classification or categories of Resources of Evidence- Based Health Care**:

• Information resources used for EBP can be classified as primary source or secondary source:

➢ **Primary sources:**

• A primary source is anything that gives you direct evidence about the people, events, or phenomena that you are researching.

• If you are researching the past, you need primary sources that were produced at the time by participants or witnesses (e.g. letters, photographs, newspapers).

• If you are researching something current, your primary sources can either be qualitative or quantitative data that you collect yourself (e.g. through interviews, surveys, experiments).

• Examples of primary sources: **Novel.** Primary sources essay by a philosopher.

2. Secondary sources:

A secondary source is anything that describes, interprets, evaluates, or analyzes information from primary sources.

• Common examples include: Books, articles and documentaries that synthesize information on a topic Summarize the original researches and include sources like systematic reviews and clinical practice guidelines.

• Examples of secondary source:
• Article analyzing the novel, textbook summarizing the philosopher‘s ideas.

Other Classification for Resources of Evidence-Based Health Care:

1-The scientific literature:

The first source of evidence is scientific research published in academic journals.

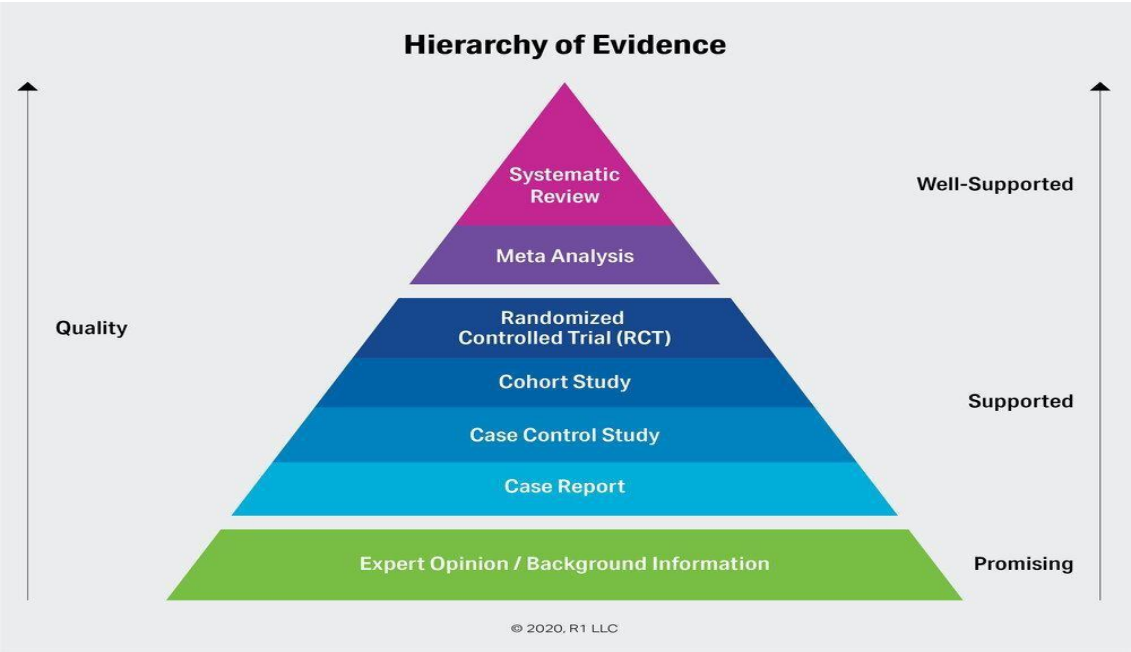
2- Evidence from the organization:

A second source of evidence is the organization itself. Whether this is a business, hospital or governmental agency.

3- Evidence from practitioners:

A third source of evidence is the professional experience such as managers, consultants, business leaders and other practitioners.

4- Evidence from stakeholders: A fourth source of evidence is stakeholder values and concerns. Stakeholders are any individuals or groups who may be affected by an organization‘s decisions.



- **Types of Evidence:**

➢ **Strongest Types of Evidence:**

1- Systematic Review:

- A systematic review is a review in which there is a comprehensive and rigorous search for relevant studies on a specific topic, and those identified are then appraised and synthesized.

- Usually focus on a clinical topic and answer a specific question.
- Searches the literature to find valid studies with good methodology.
- Systematic review is conducted by <u>systematically</u> searching the literature with strategies outlined in the methods section of the paper.
- It might list the keywords and databases used.

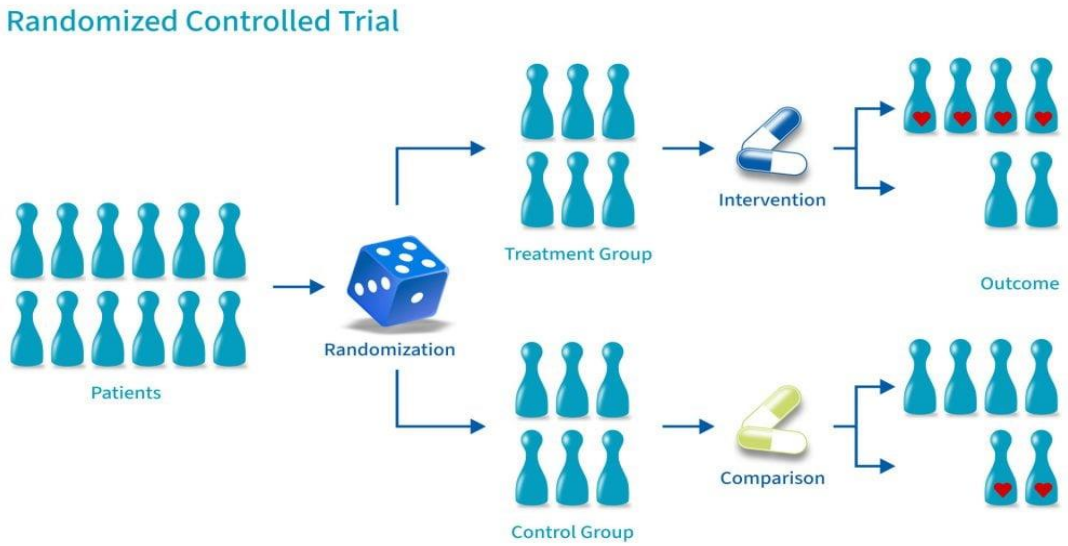
- Might also refer to limits such as years searched, languages of publications, etc...

2- Meta-Analysis:

- Meta analysis is a statistical method that combines the results of multiple studies on a certain research question.
- It is meant to give a summary of all the evidence on a certain topic, which can be used to come to more accurate and trustworthy conclusions than individual studies alone.

3- Randomized Controlled Trials:

- Research study in which participants are randomly placed in a control and comparison group. Blinding of participants and researchers is an important aspect of reliable trials. Reduces bias.



Weaker Types of Evidence:

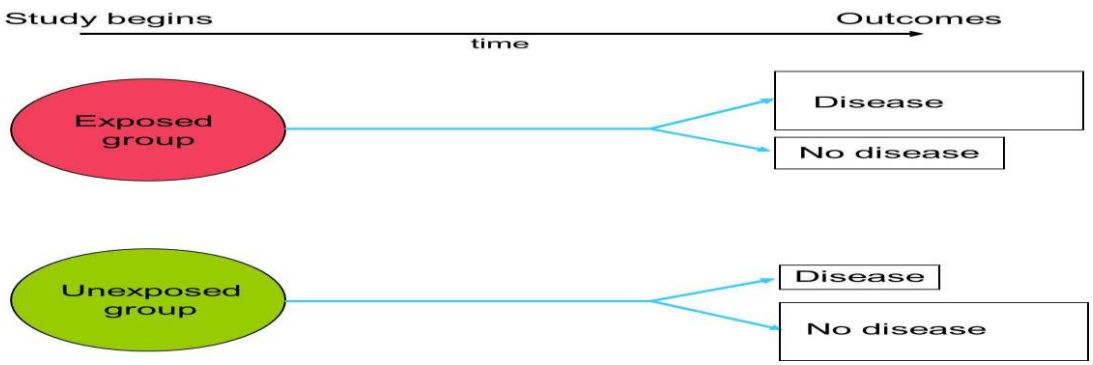
1- Cohort Studies:

- From a large population, follows patients who have a specific condition or receive a particular treatment over time and compared with another group that has not been affected by the condition or treatment studies.
- (Example: Breast Cancer patient treated with a certain type of therapy and followed over time to determine if it is effective and compared to

other breast cancer patients that did not receive this therapy and followed over time to see if there were any differences).

• This is an observational study, so it is not as reliable as a Randomized Controlled Trial due to the fact that other factors can affect group.

• Cohort study used to answer questions about etiology or prognosis. Prognostic cohort studies start with a group of patients with a specific condition and follow them up over time to see how the condition develops.



2- Case Control Studies:

Compare people who have the disease against those who do not have the disease (Example: Lung cancer studies who exposure to smoking).

3- Case Series/Case Report:

❑ Case report is a detailed report of the diagnosis, treatment, Response to treatment, and follow-up after treatment of an individual patient.

❑ Treatment of individual patients or a report on a single patient (no control group for comparison).

❑ This is low evidence because it is only referring to one patient so not as reliable as other types of research.

❑ This may, however, be all that is available. (Example: First cases of AIDS were case reports of people experiencing symptoms, etc...).

Other Types of evidence based practice resources:

1. E-Books and Libraries.

2. Electronic Databases & Article Databases.

Examples of electronic include: PubMed (MEDLINE):-

3- Internet Resources.

❖ **Required skills for locating resources:**

1.Completing independent learning modules or tutorials.

2. Attending classes about EBP searching skills.

3. Consulting with experts in EBP field or nursing field.

☒ **Systematic review**:

Introduction:

➢ A systematic review is a comprehensive and detailed approach to gathering and analyzing existing research evidence on a specific topic or research question.

➢ It is a scientific method that aims to provide an unbiased and reliable summary of available evidence to inform decision making, policy development, and further research.

➢ Systematic reviews play a crucial role in evidence-based practices as they consolidate research findings to inform decision-making. However, it is essential to assess the quality of systematic reviews to prevent biased or inaccurate conclusions.

Definition of Systematic review:

➢ A systematic review is a synthesis of the evidence on a clearly presented topic using critical methods to identify, define and assess research on the topic. A systematic review extracts and interprets data from published studies on the topic, then analyzes, describes, critically appraises, and summarizes interpretations into a refined evidence-based conclusion.

➢ **Purpose of systematic reviews:**

- Provide a comprehensive overview of the current evidence on a topic.
- Provide an unbiased and reliable assessment of the current state of knowledge.
- It allows to gather information from multiple independent studies, which increases the reliability and generalizability of the findings.
- It help healthcare professionals make evidence-based decisions.
- They allow professionals to evaluate the effectiveness of interventions, based on a large body of evidence.
- Help identify gaps in the literature, highlighting areas that require further research.

Characteristics of systematic review:

- **A systematic review has characteristics that include:**
- It involves use of clear objectives and questions that need to be answered.
- Research is found via an exhaustive search for both published and unpublished research.
- Many variables, including technique and validity, are evaluated for the included studies.
- Data from included studies may be extracted and analyzed.
- Methodologies and searches are transparent and documented; protocols may or may not be published.

❖ **Advantages of systematic reviews:**

- Reduce bias or cherry picking.
- Reliable And Accurate Results.
- Comprehensive And Exhaustive.
- Replicability.
- Identify gaps in current research.
- Supports evidence-based practice.

➢ **Limitation of systematic reviews:**

1. They encounter the same pitfalls as primary research.

a. Repeatability. b. Bias (publication).

c. Appropriate uses of methodology and statistics.

2. Location and selection of studied.

3. Inappropriate subgroup analyses.

4. Loss of information on important outcome.

5. Considered as being a wasteful use of time and resources if poorly done.

❖ **Challenges of systematic reviews:**

1– Poor access to the published literature.

2– Inadequate human resources in research.

3– Scarce local/regional funding.

4– Limited capacity for knowledge translation and dissemination.

☒ **Meta-Analysis:**

Definition:

A meta-analysis is the statistical pooling of results from several studies to generate a summary estimate of effects (e.g. from a treatment or a diagnostic tool).

✓ **The aims and goal of meta-analyses:**

1- To arrive at more accurate and reliable findings about a treatment's effect.

2- To deal with controversy when individual studies disagree.

3- To answer new questions not previously posed in component studies.

4- Explore differences between studies.

5- Generate hypotheses.

❖ **When should do a meta-analysis?**

- When more than one study has provided results about the same question.
- When there are minimal differences across studies.
- When the same outcome has been measured.

The strengths of meta-analysis:

1- Meta-analyses overcome the limitation of small sample sizes or rare outcomes by pooling results from a number of individual studies.

2- Increase the statistical power to study effects of interest.

3- The probability of perceived ―negative‖ results is reduced.

4- They can also contribute to the generalizability of study results.

❖ **Limitations of meta-analysis:**

1- Difficult and time consuming to identify appropriate studies.

2- Poor quality studies were included, increasing the risk of bias.

3- Not all studies provide adequate data for inclusion and analysis.

4- Not suitable when the studies are not similar enough.

5- Require advanced statistical techniques.